1.CPU/REGISTER ORGANIZATION

In CPU mainly we have three units:

1).ALU: ALU is called as arithmetic and logical unit which performs all the arithmetic and logical operations.

2).REGISTERS:

1 .Working registers:

a .Accumulator: This is also called as A Registers.These registers are used in addition and subtraction ,logical operation and data move instruction are given by these accumulators.

b. B-registers: used for multiplication and division

c. General purpose working registers(R0 toR7):This microcontrollers provides 8 General purpose working registers 8bit each. These are arranged as 4 selectable bank each of 8 registers. Only one bank can be active at a time. Why we use selectable bank because if we save data as R0 to R7 to ram is is difficult during interrupts but if we save data in bank during interrupt it can switch easily to another bank.

2.Memory Registers:

a. DPTR- It is called as data pointer .Made up of two8 bit registers used to fetch data from flash memory and XRAM.

b. SP- It is called as stack pointer . holds the memory address of top of the stack. It performs automatic increment when we PUSH item and automatic decrement when we call a function or POP item.

3. Status and control:

a.PSW- It is call as program status word and the status flag register . It as carry flag, overflow flag ,zero flag, Parity Flag, register bank selector bit.

4.SFR- It is called special function registers. These are registers used to control and monitor microcontroller internal function and peripherals like GPIO,I/O ports ,timer, UART,PWM and ADC registers.

3). Control unit: In control unit we have instructions ,decoder,timing,control.

- **In CPU all these are connected with internal bus.**
- **We also have RAM, ROM/Flash,peripherials like GPIO,timer,UART,ADC,etc.**

2.PROGRAM COUNTER

- **It is a register in a CPU that keeps track of the address of the next instruction that is to be executed.**
- **It is a 16 bit Register.**
- **PC points to program memory where instructions are stored .**
- **PC changes depends on the size of instruction being executed.**
- **When we use jump instruction the PC doesnot continue with the next set of instruction rather PC is loaded with the address the jump instruction specifies.**
- **A loop repeatedly executes a a particular section when a jump instruction changes the PC back to an earlier memory address.**
- **In function the return address is saved in stack, when the function is called PC address is changed with the address of the function and after the function ends the PC address returns to the main function.**
- **When an interrupt occurs the execution is temporarily stopped and moves to interrupt service routine(ISR) but the CPU must remember where the execution as stopped and this is saved in the stack. Then the PC is changed to the address of ISR.**